

# MINIMAX ALGORITHM

## AIM

To implement the **Minimax Algorithm** for game playing in Python to determine the optimal move for a player.

## ALGORITHM

1. Start.
2. Define the terminal node values.
3. If a leaf node is reached, return its value.
4. If it is the Maximizer's turn, choose the maximum value.
5. If it is the Minimizer's turn, choose the minimum value.
6. Repeat until the root node is evaluated.
7. Display the optimal value.
8. Stop.

## SAMPLE CODE

```
# Minimax Algorithm for Gaming

def minimax(depth, nodeIndex, isMax, values, height):

    if depth == height:

        return values[nodeIndex]

    if isMax:

        return max(

            minimax(depth + 1, nodeIndex * 2, False, values, height),

            minimax(depth + 1, nodeIndex * 2 + 1, False, values, height)

        )

    else:

        return min(

            minimax(depth + 1, nodeIndex * 2, True, values, height),

            minimax(depth + 1, nodeIndex * 2 + 1, True, values, height)

        )

# Terminal node values

values = [3, 5, 2, 9, 12, 5, 23, 23]

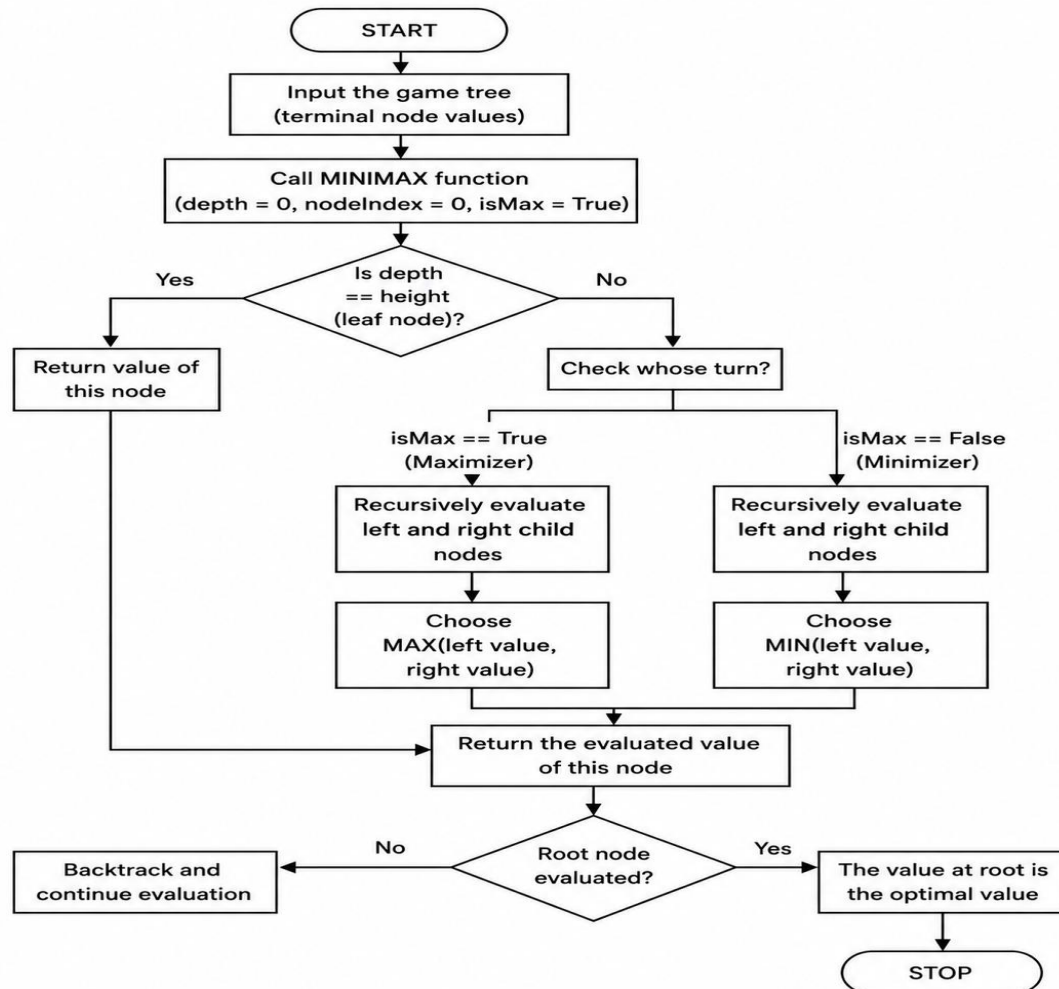
# Height of the game tree

height = 3
```

```
result = minimax(0, 0, True, values, height)
```

```
print("The optimal value is:", result)
```

## FLOW CHART



## OUTPUT

```
"C:\Users\Madhu Bavana\PycharmProjects\PythonProject1\.venv\Scripts\python.exe" "C:\Users\Madhu Ba
The optimal value is: 12

Process finished with exit code 0
```